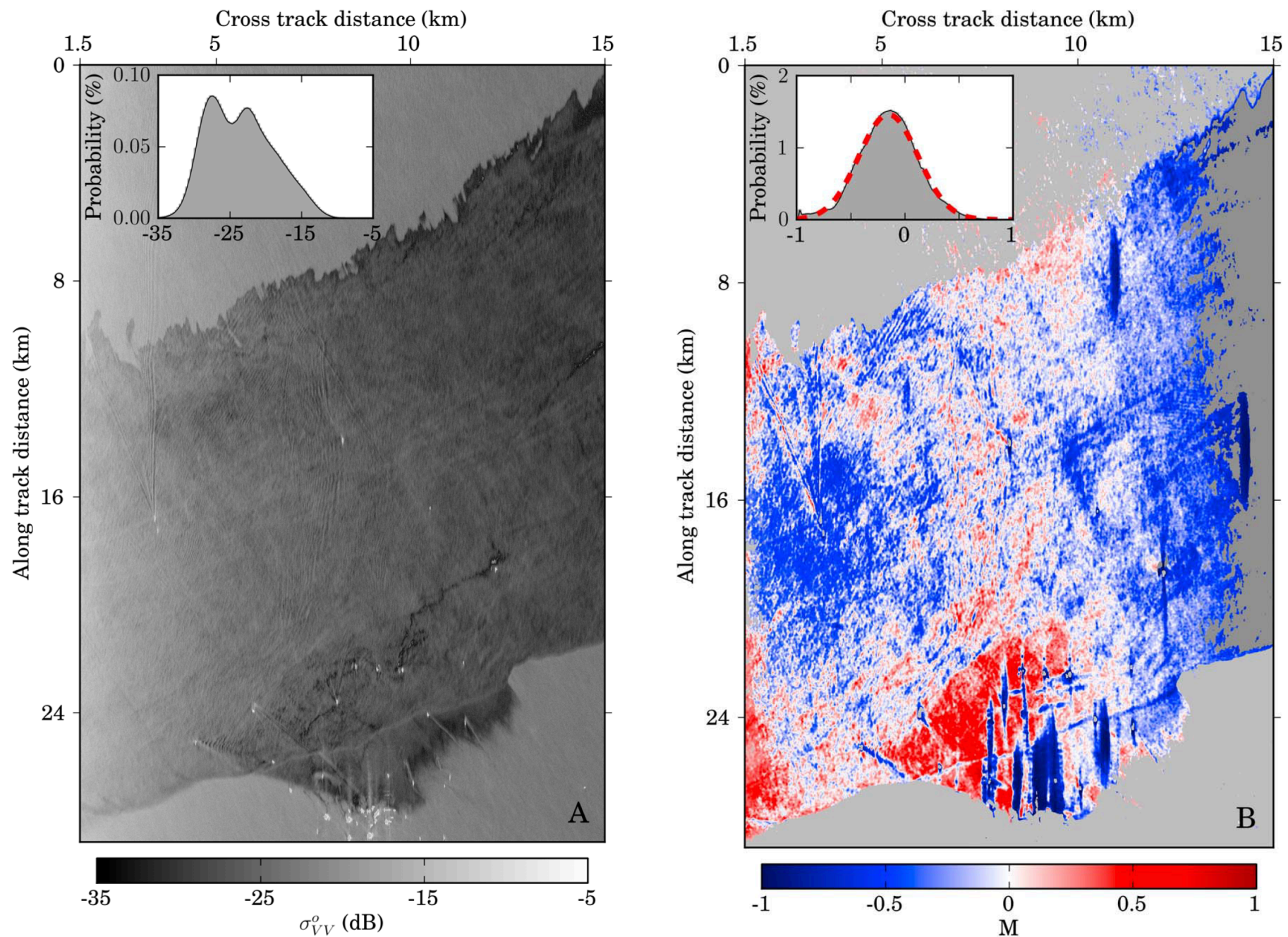


Lecture 6: Wave reflection and transmission at planar interfaces — Part 3. Multiple layers

Physical principles of remote sensing
(Ge 158)
Fall 2026
Brent Minchew

Remote sensing image of the day

$$\sigma_{pp}^0 = 4\pi k_r^4 \cos^4 \theta_f \Gamma_{pp} W \quad \Gamma_{pp} = \left| \left(\frac{\sin(\theta_i + \psi) \cos \zeta}{\sin \theta_f} \right)^2 \rho_{pp} + \left(\frac{\sin \zeta}{\sin \theta_f} \right)^2 \rho_{qq} \right|^2$$



Minchew, GRL, 2012

Topics covered

- Review of EM boundary conditions
- Review of 3 laws of geometric optics
 - Plane of incidence
 - Law of reflection
 - Law of refraction → total internal reflection
- Scattering from a thin layer at oblique incidence; serves as;
 - Powerful model for various applications
 - Review of oblique reflection and transmission
 - Foundation for more sophisticated surface scattering models

Review

- Incidence to a planar boundary
 - (Angular) frequency is constant across the boundary (necessary for continuity)
 - Reflection and transmission are a functions of the relative phase velocities (i.e., wavenumbers and wavelengths) in the media; true for all wave types, not just EM waves
 - For EM waves, reflection and transmission depend on square roots of dielectric constants (i.e., indices of refraction)
- Boundary conditions (BCs) for EM waves can be derived from Maxwell's Equations. For RS applications,
 - Component of electric displacement ($\epsilon \mathbf{E}$) normal to interface is constant
 - Component of ($\mu \mathbf{H}$) field normal to interface is constant (generally the case that $\mu_1 = \mu_2 = \mu_0$)
 - Components of $\mathbf{E}$ field tangent to the interface are constant
 - Components of $\mathbf{H}$ field tangent to the interface are constant
- 3 laws of geometric optics
 - Plane of incidence (in class, this is the chalkboard) contains incident, reflected, and transmitted rays along with interface normal
 - Incidence angle = angle of reflection
 - Snell's Law: $\sqrt{\epsilon_1} \sin \theta_1 = \sqrt{\epsilon_2} \sin \theta_2$
- Applying BCs to EM wave gives Fresnel Equations for oblique incidence

$$\rho_h = \frac{E_{h0}^r}{E_{h0}^i} = \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2} \quad \tau_h = \frac{E_{h0}^t}{E_{h0}^i} = \frac{2\eta_2 \cos \theta_1}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2}$$

$$\rho_v = \frac{E_{v0}^r}{E_{v0}^i} = \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1} = \frac{-\frac{\epsilon_2}{\epsilon_1} \cos \theta_1 + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_1}}{\frac{\epsilon_2}{\epsilon_1} \cos \theta_1 + \sqrt{\frac{\epsilon_2}{\epsilon_1} - \sin^2 \theta_1}} \quad \tau_v = \frac{E_{v0}^t}{E_{v0}^i} = (1 + \rho_v) \frac{\cos \theta_1}{\cos \theta_2}$$

Effective reflection from two-layer composite

$$\rho_p = \frac{\rho_{12p} + \rho_{23p} \exp(-2\gamma_2 d \cos \theta_2)}{1 + \rho_{12p} \rho_{23p} \exp(-2\gamma_2 d \cos \theta_2)}$$

where ρ_{12p} and ρ_{23p} are the Fresnel reflection coefficients for p ($=h$ or v) polarization at each interface

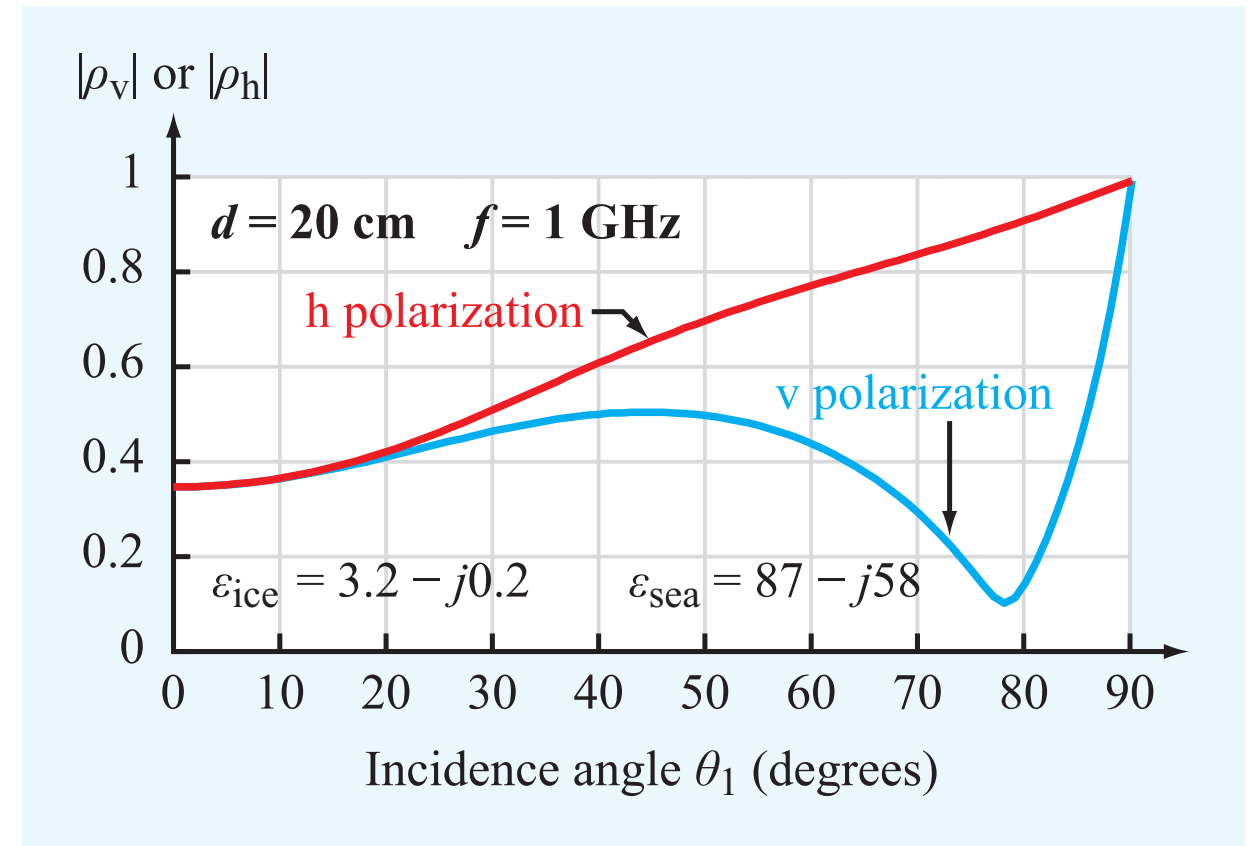
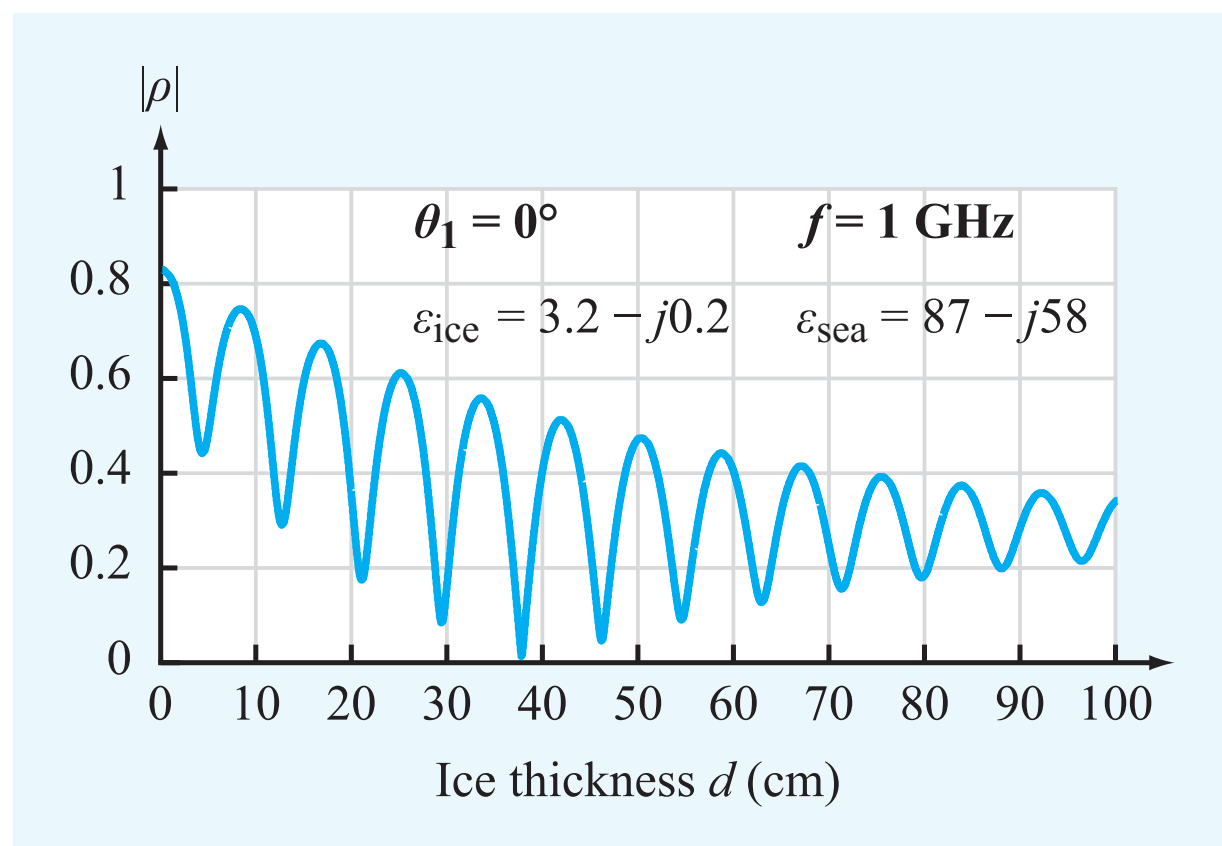
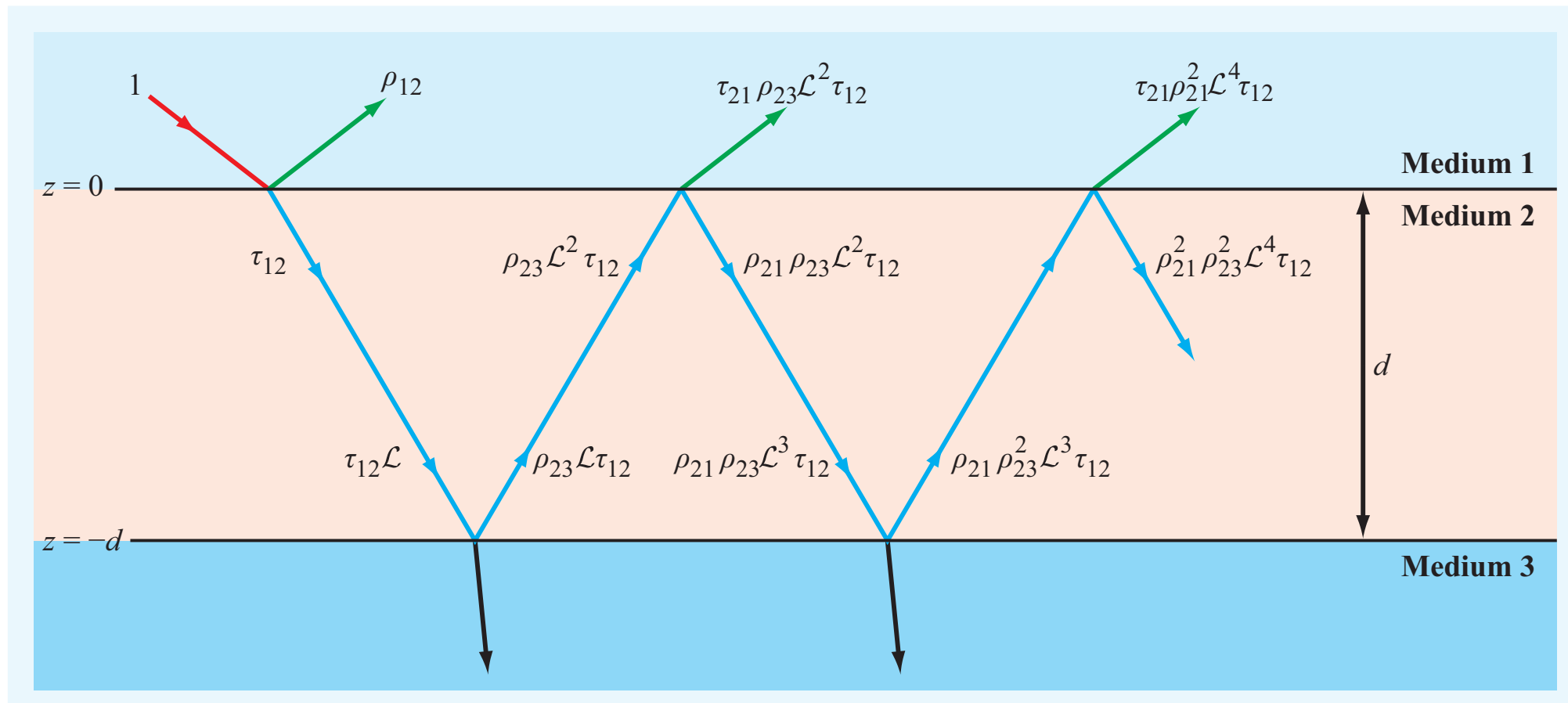


Figure 2-22: Plots of (a) $|\rho|$ as a function of ice-layer thickness d for normal incidence and (b) $|\rho_v|$ and $|\rho_h|$ as a function of incidence angle θ_1 for a 20 cm ice layer over the ocean surface.

Summary (derivation in class)



- Effective reflection coefficient from a 2-layer composite

$$\rho_p = \frac{\rho_{12p} + \rho_{23p} \exp(-2\gamma_2 d \cos \theta_2)}{1 + \rho_{12p} \rho_{23p} \exp(-2\gamma_2 d \cos \theta_2)}$$

$$\rho_{12h} = \frac{\eta_2 \cos \theta_1 - \eta_1 \cos \theta_2}{\eta_2 \cos \theta_1 + \eta_1 \cos \theta_2}$$

$$\rho_{12v} = \frac{\eta_2 \cos \theta_2 - \eta_1 \cos \theta_1}{\eta_2 \cos \theta_2 + \eta_1 \cos \theta_1}$$

$$\rho_{23h} = \frac{\eta_3 \cos \theta_2 - \eta_2 \cos \theta_3}{\eta_3 \cos \theta_2 + \eta_2 \cos \theta_3}$$

$$\rho_{23v} = \frac{\eta_3 \cos \theta_3 - \eta_2 \cos \theta_2}{\eta_3 \cos \theta_3 + \eta_2 \cos \theta_2}$$